

# Normal Distribution

## Key Notes and Formulae

### 1. What is a normal distribution?

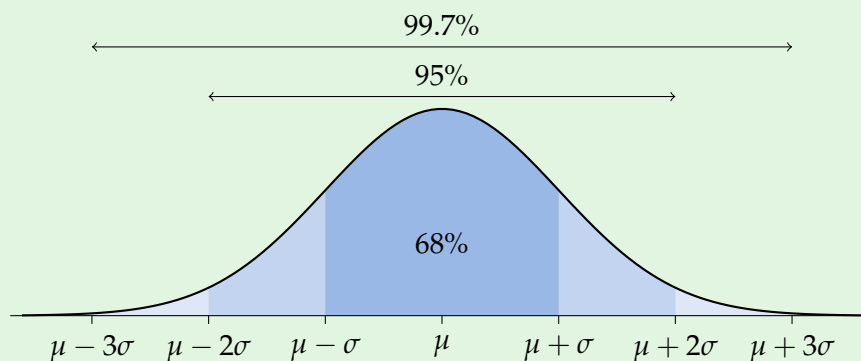
A **continuous** random variable  $X$  (heights, weights, times, ...) whose graph is a **bell-shaped** curve.

We write  $X \sim N(\mu, \sigma^2)$  where  $\mu$  = mean and  $\sigma^2$  = **variance** ( $\sigma$  = standard deviation).

**Careful:**  $N(20, 9)$  means  $\sigma^2 = 9$ , so  $\sigma = 3$ .

- The curve is **symmetric** about the mean: mean = median = mode =  $\mu$ .
- The total area under the curve is 1, so  $P(X < \mu) = P(X > \mu) = 0.5$ .
- Probability = **area** under the curve. For a continuous variable  $P(X = a) = 0$ , so  $P(X < a) = P(X \leq a)$ .

### 2. The 68 – 95 – 99.7 rule (approximately)



### 3. Standardising – the z-score

$$z = \frac{x - \mu}{\sigma} \iff x = \mu + z\sigma, \quad Z \sim N(0, 1^2)$$

$z$  tells you **how many standard deviations**  $x$  is above ( $z > 0$ ) or below ( $z < 0$ ) the mean.  
e.g.  $\mu = 100, \sigma = 10$ : the value 1.12 s.d. above the mean is  $x = 100 + 1.12(10) = 111.2$ .

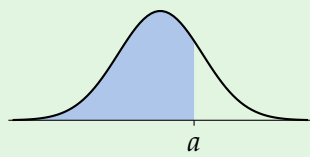
### 4. Using the GDC

|                                                            |                                                                                                                                                                           |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>You know <math>x</math>,<br/>want a probability</b>     | $P(a < X < b) = \text{normalcdf}(a, b, \mu, \sigma)$<br>For “less than” use lower = $-10^{99}$ ; for “more than” use upper = $10^{99}$ .                                  |
| <b>You know a<br/>probability,<br/>want <math>x</math></b> | $x = \text{invNorm}(\text{area}, \mu, \sigma)$ – the area is to the <b>left</b> of $x$ .<br>If $P(X > a) = 0.27$ , use area = $1 - 0.27 = 0.73$ (or choose “tail right”). |

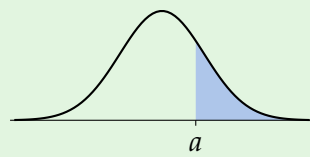
Always state  $X \sim N(\mu, \sigma^2)$  and write the probability statement before the answer.

**5. Always sketch!**

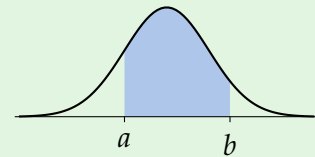
Draw the curve, mark the mean in the middle, mark the value(s), and shade the region you want.



$$P(X < a)$$



$$P(X > a) = 1 - P(X < a)$$



$$P(a < X < b)$$

**6. Using symmetry** (no calculator needed)

- $P(X < \mu - k) = P(X > \mu + k)$  – the two tails the same distance from  $\mu$  are equal.
- If  $P(X < d) = P(X > 105)$  and  $\mu = 100$ , then  $d = 100 - 5 = 95$ .
- $P(\mu < X < \mu + k) = 0.5 - P(X > \mu + k)$ . e.g.  $P(X > 24) = 0.06$ ,  $\mu = 17 \Rightarrow P(17 < X < 24) = 0.44$ .

**7. Finding an unknown mean  $\mu$  or standard deviation  $\sigma$** 

1. Find  $z$  from the **standard** normal:  $z = \text{invNorm}(\text{area to the left}, 0, 1)$ .

2. Substitute into  $z = \frac{x - \mu}{\sigma}$  and solve for the unknown.

e.g.  $\sigma = 5$ ,  $P(X < 153) = 0.705$ :  $z = \text{invNorm}(0.705, 0, 1) = 0.5388$ , so  $0.5388 = \frac{153 - \mu}{5} \Rightarrow \mu = 150$ .

**Both  $\mu$  and  $\sigma$  unknown:** use two pieces of information to make **two equations** and solve simultaneously:

$$x_1 = \mu + z_1\sigma, \quad x_2 = \mu + z_2\sigma$$

**8. Normal + binomial together**

First use the normal distribution to find the probability  $p$  of “success” for **one** item.

Then for  $n$  items use  $Y \sim B(n, p)$ .

e.g. all 10 chickens:  $p^{10}$ ; at least 7:  $\text{binomCdf}(10, p, 7, 10)$ .

**Two** items both in a range:  $p \times p = p^2$ .

**9. Conditional probability (tree diagrams)**

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad \text{e.g.} \quad P(\text{girl} | \text{tall}) = \frac{P(\text{girl}) \times P(\text{tall} | \text{girl})}{P(\text{tall})}$$

**How to use this worksheet:** questions go from easy to difficult. Attempt each page first, then check the yellow answer page that follows it.

**Question 1** Non-calculator Easy

[Maximum mark: 6]

A random variable  $X$  is distributed normally with a mean of 100 and a variance of 100.

(a) Find the value of  $X$  that is 1.12 standard deviations **above** the mean. [4]

(b) Find the value of  $X$  that is 1.12 standard deviations **below** the mean. [2]

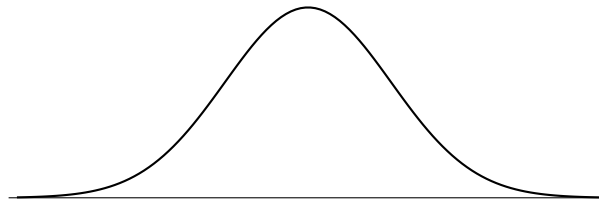
Working:

**Question 2** Non-calculator Easy

[Maximum mark: 6]

Let  $X$  be normally distributed with mean 100 cm and standard deviation 5 cm.

(a) On the diagram below, shade the region representing  $P(X > 105)$ . [2]



(b) Given that  $P(X < d) = P(X > 105)$ , find the value of  $d$ . [2]

(c) Given that  $P(X > 105) = 0.16$  (correct to two significant figures), find  $P(d < X < 105)$ . [2]

Working:

## Answers – Questions 1 & 2

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### Question 1

Variance = 100, so  $\sigma = \sqrt{100} = 10$ .

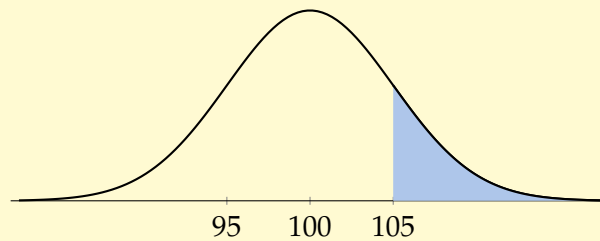
(a)  $x = \mu + 1.12\sigma = 100 + 1.12 \times 10 = 100 + 11.2 = \boxed{111.2}$

(or  $\frac{x - 100}{10} = 1.12 \Rightarrow x = 111.2$ )

(b)  $x = \mu - 1.12\sigma = 100 - 11.2 = \boxed{88.8}$

### Question 2

(a) Shade the area to the right of 105.



(b) By symmetry, 105 is 5 above the mean, so  $d$  is 5 below the mean:  $d = 100 - 5 = \boxed{95}$

(c)  $P(X < 95) = P(X > 105) = 0.16$ , so

$$P(95 < X < 105) = 1 - 0.16 - 0.16 = \boxed{0.68}$$

**Question 3** Calculator Easy

[Maximum mark: 6]

A random variable  $X$  is distributed normally with mean 450 and standard deviation 20.

(a) Find  $P(X \leq 475)$ . [2]

(b) Given that  $P(X > a) = 0.27$ , find  $a$ . [4]

Working:

**Question 4** Calculator Easy

[Maximum mark: 6]

The heights of a group of students are normally distributed with a mean of 160 cm and a standard deviation of 20 cm.

(a) A student is chosen at random. Find the probability that the student's height is greater than 180 cm.

(b) In this group of students, 11.9% have heights less than  $d$  cm. Find the value of  $d$ .

Working:

## Answers – Questions 3 & 4

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### Question 3

$$X \sim N(450, 20^2)$$

$$(a) P(X \leq 475) = \text{normalcdf}(-10^{99}, 475, 450, 20) \approx \boxed{0.894}$$

$$(b) P(X > a) = 0.27 \Rightarrow P(X < a) = 1 - 0.27 = 0.73$$

$$a = \text{invNorm}(0.73, 450, 20) \approx \boxed{462} \quad (462.26)$$

### Question 4

$$H \sim N(160, 20^2)$$

$$(a) P(H > 180) = \text{normalcdf}(180, 10^{99}, 160, 20) \approx \boxed{0.159} \quad (180 \text{ is exactly } 1\sigma \text{ above the mean})$$

$$(b) P(H < d) = 0.119$$

$$d = \text{invNorm}(0.119, 160, 20) \approx \boxed{136}$$

**Question 5** Calculator Easy

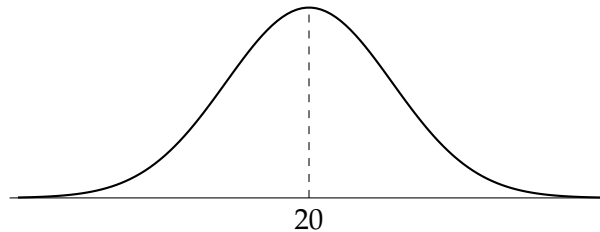
[Maximum mark: 8]

A random variable  $X$  is distributed normally with a mean of 20 and variance 9.

(a) Find  $P(X \leq 24.5)$ . [3]

(b) Let  $P(X \leq k) = 0.85$ .

(i) Represent this information on the following diagram.



(ii) Find the value of  $k$ . [5]

Working:

**Question 6** Calculator Easy

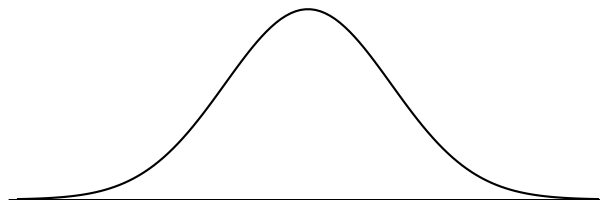
[Maximum mark: 6]

The weights of a group of children are normally distributed with a mean of 22.5 kg and a standard deviation of 2.2 kg.

(a) Write down the probability that a child selected at random has a weight more than 25.8 kg.

(b) Of the group 95% weigh less than  $k$  kilograms. Find the value of  $k$ .

(c) The diagram below shows a normal curve. On the diagram, shade the region that represents the following information: *87% of the children weigh less than 25 kg.*



Working:

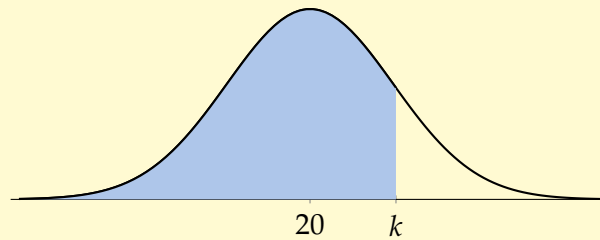
## Answers – Questions 5 & 6

### Question 5

$X \sim N(20, 9)$ , so  $\sigma = \sqrt{9} = 3$ .

(a)  $P(X \leq 24.5) = \text{normalcdf}(-10^{99}, 24.5, 20, 3) \approx \boxed{0.933}$

(b) (i)  $k$  is to the right of the mean; shade the area to the left of  $k$  (85%).



(ii)  $k = \text{invNorm}(0.85, 20, 3) \approx \boxed{23.1}$

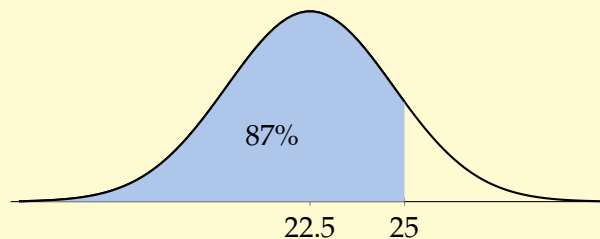
### Question 6

$W \sim N(22.5, 2.2^2)$

(a)  $P(W > 25.8) = \text{normalcdf}(25.8, 10^{99}, 22.5, 2.2) \approx \boxed{0.0668}$

(b)  $k = \text{invNorm}(0.95, 22.5, 2.2) \approx \boxed{26.1 \text{ kg}}$

(c) Vertical line at 25 (to the right of the mean), shade everything to the left of it.



**Question 7** Calculator Medium

[Maximum mark: 8]

Residents of a small town have savings which are normally distributed with a mean of \$3000 and a standard deviation of \$500.

- (i) What percentage of townspeople have savings greater than \$3200?
- (ii) Two townspeople are chosen at random. What is the probability that **both** of them have savings between \$2300 and \$3300?
- (iii) The percentage of townspeople with savings less than  $d$  dollars is 74.22%. Find the value of  $d$ .

Working:

**Question 8** Non-calculator Medium

[Maximum mark: 9]

The heights of trees in a forest are normally distributed with mean height 17 metres. One tree is selected at random. The probability that a selected tree has a height greater than 24 metres is 0.06.

- (a) Find the probability that the tree selected has a height less than 24 metres. [2]
- (b) The probability that the tree has a height less than  $D$  metres is 0.06. Find the value of  $D$ . [3]
- (c) A woodcutter randomly selects 200 trees. Find the expected number of trees whose height lies between 17 metres and 24 metres. [4]

Working:

## Answers – Questions 7 & 8

### Question 7

$$S \sim N(3000, 500^2)$$

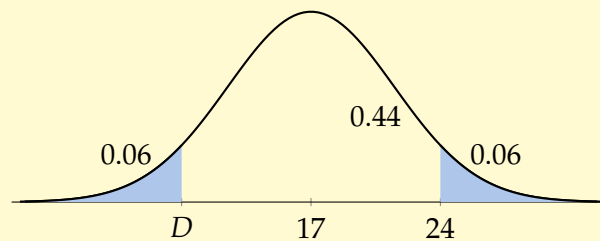
$$(i) P(S > 3200) = \text{normalcdf}(3200, 10^{99}, 3000, 500) \approx 0.345 = \boxed{34.5\%}$$

$$(ii) P(2300 < S < 3300) = \text{normalcdf}(2300, 3300, 3000, 500) \approx 0.645$$

$$P(\text{both}) = (0.645)^2 \approx \boxed{0.416}$$

$$(iii) d = \text{invNorm}(0.7422, 3000, 500) \approx \boxed{\$3325} \quad (\$3330 \text{ to 3 s.f.})$$

### Question 8



$$(a) P(H < 24) = 1 - 0.06 = \boxed{0.94}$$

$$(b) \text{ By symmetry, 24 is 7 m above the mean, so } D \text{ is 7 m below the mean: } D = 17 - 7 = \boxed{10}$$

$$(c) P(17 < H < 24) = 0.5 - 0.06 = 0.44$$

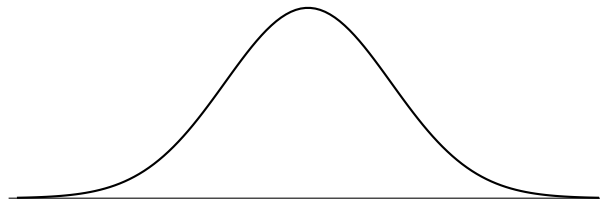
$$\text{Expected number} = 200 \times 0.44 = \boxed{88 \text{ trees}}$$

**Question 9** Calculator Medium

[Maximum mark: 13]

A box contains a large number of biscuits. The weights of biscuits are normally distributed with mean 7 g and standard deviation 0.5 g.

- (a) One biscuit is chosen at random from the box. Find the probability that this biscuit
- (i) weighs less than 8 g;
  - (ii) weighs between 6 g and 8 g. [4]
- (b) Five percent of the biscuits in the box weigh less than  $d$  grams.
- (i) Complete the following normal distribution diagram, to represent this information, by indicating  $d$ , and shading the appropriate region.



- (ii) Find the value of  $d$ . [5]
- (c) The weights of biscuits in another box are normally distributed with mean  $\mu$  and standard deviation 0.5 g. It is known that 20% of the biscuits in this second box weigh less than 5 g. Find the value of  $\mu$ . [4]

Working:

## Answers – Question 9

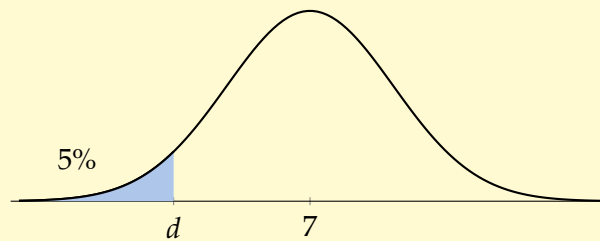
### Question 9

$$X \sim N(7, 0.5^2)$$

(a) (i)  $P(X < 8) = \text{normalcdf}(-10^{99}, 8, 7, 0.5) \approx \boxed{0.977}$

(ii)  $P(6 < X < 8) = \text{normalcdf}(6, 8, 7, 0.5) \approx \boxed{0.954}$  (within  $2\sigma$  of the mean,  $\approx 95\%$ )

(b) (i)  $d$  to the left of the mean, with the small area to the left of  $d$  shaded.



(ii)  $d = \text{invNorm}(0.05, 7, 0.5) \approx \boxed{6.18 \text{ g}}$

(c)  $Y \sim N(\mu, 0.5^2)$ ,  $P(Y < 5) = 0.2$

$$z = \text{invNorm}(0.2, 0, 1) = -0.8416$$

$$\frac{5 - \mu}{0.5} = -0.8416 \Rightarrow \mu = 5 + 0.5(0.8416) \approx \boxed{5.42 \text{ g}}$$

**Question 10** Calculator Medium

[Maximum mark: 6]

The heights of boys at a particular school follow a normal distribution with a standard deviation of 5 cm. The probability of a boy being shorter than 153 cm is 0.705.

- (a) Calculate the mean height of the boys.
- (b) Find the probability of a boy being taller than 156 cm.

Working:

**Question 11** Calculator Medium

[Maximum mark: 7]

The scores of a test given to students are normally distributed with a mean of 21. 80% of the students have scores less than 23.7.

- (a) Find the standard deviation of the scores. [3]

A student is chosen at random. This student has the same probability of having a score less than 25.4 as having a score greater than  $b$ .

- (b) (i) Find the probability the student has a score less than 25.4.
- (ii) Find the value of  $b$ . [4]

Working:

## Answers – Questions 10 & 11

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### Question 10

$$H \sim N(\mu, 5^2), P(H < 153) = 0.705$$

$$(a) z = \text{invNorm}(0.705, 0, 1) = 0.5388$$

$$\frac{153 - \mu}{5} = 0.5388 \Rightarrow \mu = 153 - 5(0.5388) = 150.31 \approx \boxed{150 \text{ cm}}$$

$$(b) P(H > 156) = \text{normalcdf}(156, 10^{99}, 150.31, 5) \approx \boxed{0.127} \quad (0.128 \text{ if } \mu = 150.3 \text{ is used})$$

### Question 11

$$X \sim N(21, \sigma^2)$$

$$(a) P(X < 23.7) = 0.8 \Rightarrow z = \text{invNorm}(0.8, 0, 1) = 0.8416$$

$$\frac{23.7 - 21}{\sigma} = 0.8416 \Rightarrow \sigma = \frac{2.7}{0.8416} \approx \boxed{3.21}$$

$$(b) (i) P(X < 25.4) = \text{normalcdf}(-10^{99}, 25.4, 21, 3.208) \approx \boxed{0.915}$$

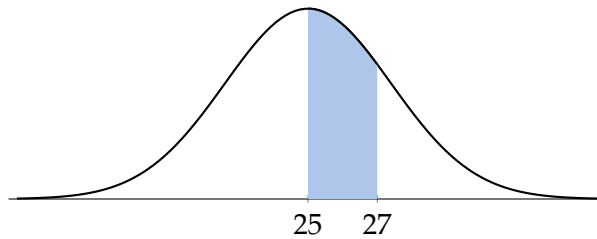
(ii) By symmetry: 25.4 is  $25.4 - 21 = 4.4$  above the mean, so  $b$  is 4.4 below the mean.

$$b = 21 - 4.4 = \boxed{16.6}$$

**Question 12** Calculator Medium

[Maximum mark: 7]

Let the random variable  $X$  be normally distributed with mean 25, as shown in the following diagram.



The shaded region between 25 and 27 represents 30% of the distribution.

- (a) Find  $P(X > 27)$ . [2]
- (b) Find the standard deviation of  $X$ . [5]

Working:

**Question 13** Calculator Medium

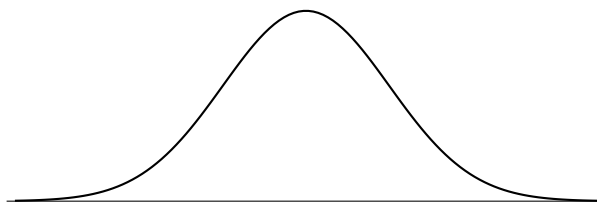
[Maximum mark: 7]

The heights of certain plants are normally distributed. The plants are classified into three categories.

The shortest 12.92% are in category A. The tallest 10.38% are in category C.

All the other plants are in category B with heights between  $r$  cm and  $t$  cm.

- (a) Complete the following diagram to represent this information. [2]



- (b) Given that the mean height is 6.84 cm and the standard deviation 0.25 cm, find the value of  $r$  and of  $t$ . [5]

Working:

## Answers – Questions 12 & 13

### Question 12

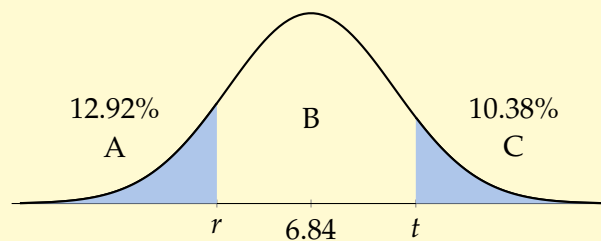
(a) Half the area is above 25, so  $P(X > 27) = 0.5 - 0.3 = \boxed{0.2}$

(b)  $P(X < 27) = 0.8 \Rightarrow z = \text{invNorm}(0.8, 0, 1) = 0.8416$

$$\frac{27 - 25}{\sigma} = 0.8416 \Rightarrow \sigma = \frac{2}{0.8416} \approx \boxed{2.38}$$

### Question 13

(a) Three regions: A on the left (12.92%), B in the middle, C on the right (10.38%).



(b)  $X \sim N(6.84, 0.25^2)$

$$r = \text{invNorm}(0.1292, 6.84, 0.25) \approx \boxed{6.56 \text{ cm}}$$

$$P(X < t) = 1 - 0.1038 = 0.8962: t = \text{invNorm}(0.8962, 6.84, 0.25) \approx \boxed{7.16 \text{ cm}}$$

**Question 14** Calculator Hard

[Maximum mark: 6]

The heights of certain flowers follow a normal distribution. It is known that 20% of these flowers have a height less than 3 cm and 10% have a height greater than 8 cm. Find the value of the mean  $\mu$  and the standard deviation  $\sigma$ .

Working:

**Question 15** Calculator Hard

[Maximum mark: 6]

The speeds of cars at a certain point on a straight road are normally distributed with mean  $\mu$  and standard deviation  $\sigma$ . 15% of the cars travelled at speeds greater than 90 km h<sup>-1</sup> and 12% of them at speeds less than 40 km h<sup>-1</sup>. Find  $\mu$  and  $\sigma$ .

Working:

## Answers – Questions 14 & 15

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### Question 14

$$X \sim N(\mu, \sigma^2), \quad P(X < 3) = 0.2 \quad \text{and} \quad P(X > 8) = 0.1 \Rightarrow P(X < 8) = 0.9$$

$$z_1 = \text{invNorm}(0.2, 0, 1) = -0.8416, \quad z_2 = \text{invNorm}(0.9, 0, 1) = 1.2816$$

$$\frac{3 - \mu}{\sigma} = -0.8416 \Rightarrow 3 = \mu - 0.8416\sigma \qquad \frac{8 - \mu}{\sigma} = 1.2816 \Rightarrow 8 = \mu + 1.2816\sigma$$

$$\text{Subtract: } 5 = 2.1232\sigma \Rightarrow \boxed{\sigma \approx 2.35}, \quad \text{then } \mu = 3 + 0.8416(2.355) \Rightarrow \boxed{\mu \approx 4.98}$$

(Solve the two equations on the GDC.)

### Question 15

$$P(X > 90) = 0.15 \Rightarrow P(X < 90) = 0.85, \quad P(X < 40) = 0.12$$

$$z_1 = \text{invNorm}(0.85, 0, 1) = 1.0364, \quad z_2 = \text{invNorm}(0.12, 0, 1) = -1.1750$$

$$90 = \mu + 1.0364\sigma \qquad 40 = \mu - 1.1750\sigma$$

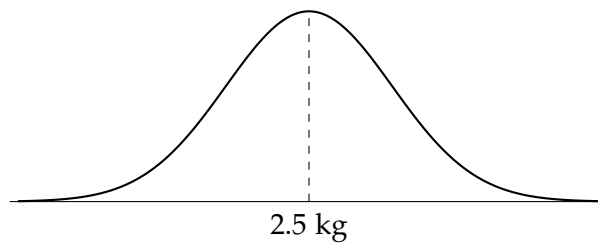
$$\text{Subtract: } 50 = 2.2114\sigma \Rightarrow \boxed{\sigma \approx 22.6}, \quad \mu = 90 - 1.0364(22.61) \Rightarrow \boxed{\mu \approx 66.6}$$

**Question 16** Calculator Hard

[Maximum mark: 13]

The weights of chickens for sale in a shop are normally distributed with mean 2.5 kg and standard deviation 0.3 kg.

- (a) A chicken is chosen at random.
- (i) Find the probability that it weighs less than 2 kg.
  - (ii) Find the probability that it weighs more than 2.8 kg.
  - (iii) Copy the diagram below. Shade the areas that represent the probabilities from parts (i) and (ii).



- (iv) **Hence** show that the probability that it weighs between 2 kg and 2.8 kg is 0.7936 (to four significant figures). [7]
- (b) A customer buys 10 chickens.
- (i) Find the probability that all 10 chickens weigh between 2 kg and 2.8 kg.
  - (ii) Find the probability that at least 7 of the chickens weigh between 2 kg and 2.8 kg. [6]

Working:

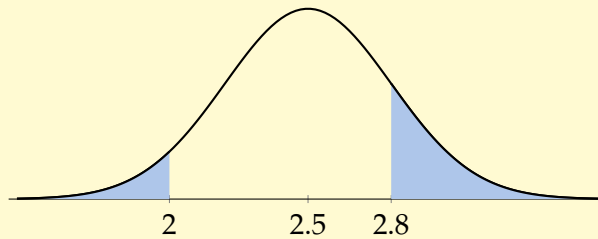
## Answers – Question 16

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### Question 16

$$W \sim N(2.5, 0.3^2)$$

- (a) (i)  $P(W < 2) = \text{normalcdf}(-10^{99}, 2, 2.5, 0.3) \approx \boxed{0.0478}$   
(ii)  $P(W > 2.8) = \text{normalcdf}(2.8, 10^{99}, 2.5, 0.3) \approx \boxed{0.159}$   
(iii) Shade to the left of 2 and to the right of 2.8.



- (iv)  $P(2 < W < 2.8) = 1 - 0.04779 - 0.15866 = 0.79355 \dots = \boxed{0.7936}$  (4 s.f.)
- (b) Let  $Y$  be the number of chickens (out of 10) weighing between 2 kg and 2.8 kg.  
 $Y \sim B(10, 0.79355)$ .
- (i)  $P(Y = 10) = (0.79355)^{10} \approx \boxed{0.0990}$   
(ii)  $P(Y \geq 7) = \text{binomCdf}(10, 0.79355, 7, 10) \approx \boxed{0.867}$

**Question 17** Calculator Hard

[Maximum mark: 13]

A van can take either Route A or Route B for a particular journey.

If Route A is taken, the journey time may be assumed to be normally distributed with mean 46 minutes and a standard deviation 10 minutes.

If Route B is taken, the journey time may be assumed to be normally distributed with mean  $\mu$  minutes and standard deviation 12 minutes.

- (a) For Route A, find the probability that the journey takes more than 60 minutes. [2]
- (b) For Route B, the probability that the journey takes less than 60 minutes is 0.85. Find the value of  $\mu$ . [3]
- (c) The van sets out at 06:00 and needs to arrive before 07:00.
  - (i) Which route should it take?
  - (ii) Justify your answer. [3]
- (d) On five consecutive days the van sets out at 06:00 and takes Route B. Find the probability that
  - (i) it arrives before 07:00 on all five days;
  - (ii) it arrives before 07:00 on at least three days. [5]

Working:

## Answers – Question 17

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### Question 17

(a)  $A \sim N(46, 10^2)$ :  $P(A > 60) = \text{normalcdf}(60, 10^{99}, 46, 10) \approx \boxed{0.0808}$

(b)  $B \sim N(\mu, 12^2)$ ,  $P(B < 60) = 0.85 \Rightarrow z = \text{invNorm}(0.85, 0, 1) = 1.0364$

$$\frac{60 - \mu}{12} = 1.0364 \Rightarrow \mu = 60 - 12(1.0364) \approx \boxed{47.6 \text{ minutes}}$$

(c) (i) **Route A.**

(ii) It must arrive within 60 minutes.  $P(A < 60) = 1 - 0.0808 = 0.919$ ,  $P(B < 60) = 0.85$ .

Since  $0.919 > 0.85$ , Route A gives a higher probability of arriving before 07:00.

(d) Let  $Y$  be the number of days (out of 5) it arrives before 07:00.  $Y \sim B(5, 0.85)$ .

(i)  $P(Y = 5) = (0.85)^5 \approx \boxed{0.444}$

(ii)  $P(Y \geq 3) = \text{binomCdf}(5, 0.85, 3, 5) \approx \boxed{0.973}$

**Question 18** Calculator Hard

[Maximum mark: 17]

In a large school, the heights of all fourteen-year-old students are measured.

The heights of the girls are normally distributed with mean 155 cm and standard deviation 10 cm.

The heights of the boys are normally distributed with mean 160 cm and standard deviation 12 cm.

- (a) Find the probability that a girl is taller than 170 cm. [3]
- (b) Given that 10% of the girls are shorter than  $x$  cm, find  $x$ . [3]
- (c) Given that 90% of the boys have heights between  $q$  cm and  $r$  cm where  $q$  and  $r$  are symmetrical about 160 cm, and  $q < r$ , find the value of  $q$  and of  $r$ . [4]

In the group of fourteen-year-old students, 60% are girls and 40% are boys.

The probability that a girl is taller than 170 cm was found in part (a).

The probability that a boy is taller than 170 cm is 0.202.

A fourteen-year-old student is selected at random.

- (d) Calculate the probability that the student is taller than 170 cm. [4]
- (e) Given that the student is taller than 170 cm, what is the probability the student is a girl? [3]

Working:

## Answers – Question 18

### Question 18

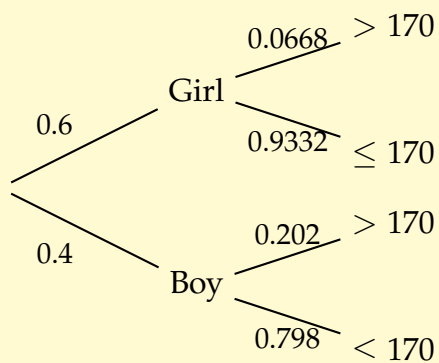
Girls  $G \sim N(155, 10^2)$ , boys  $B \sim N(160, 12^2)$

(a)  $P(G > 170) = \text{normalcdf}(170, 10^{99}, 155, 10) \approx \boxed{0.0668}$

(b)  $x = \text{invNorm}(0.1, 155, 10) \approx \boxed{142 \text{ cm}}$

(c) 90% in the middle leaves 5% in each tail.

$q = \text{invNorm}(0.05, 160, 12) \approx \boxed{140 \text{ cm}}, \quad r = \text{invNorm}(0.95, 160, 12) \approx \boxed{180 \text{ cm}}$



(d)  $P(T) = 0.6(0.0668) + 0.4(0.202)$   
 $= 0.04009 + 0.0808$   
 $= 0.12088 \approx \boxed{0.121}$

(e)  $P(\text{girl} | T) = \frac{0.6 \times 0.0668}{0.12088}$   
 $\approx \boxed{0.332}$

**Question 19** Calculator Hard

[Maximum mark: 15]

The weights of players in a sports league are normally distributed with a mean of 76.6 kg, (correct to three significant figures). It is known that 80% of the players have weights between 68 kg and 82 kg. The probability that a player weighs less than 68 kg is 0.05.

- (a) Find the probability that a player weighs more than 82 kg. [2]
- (b) (i) Write down the standardized value,  $z$ , for 68 kg.  
(ii) Hence, find the standard deviation of weights. [4]

To take part in a tournament, a player's weight must be within 1.5 standard deviations of the mean.

- (c) (i) Find the set of all possible weights of players that take part in the tournament.  
(ii) A player is selected at random. Find the probability that the player takes part in the tournament. [5]

Of the players in the league, 25% are women. Of the women, 70% take part in the tournament.

- (d) Given that a player selected at random takes part in the tournament, find the probability that the selected player is a woman. [4]

Working:

## Answers – Question 19

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### Question 19

$$W \sim N(76.6, \sigma^2)$$

$$(a) P(W > 82) = 1 - 0.80 - 0.05 = \boxed{0.15}$$

$$(b) (i) P(W < 68) = 0.05 \Rightarrow z = \text{invNorm}(0.05, 0, 1) \approx \boxed{-1.64} \quad (-1.645)$$

$$(ii) \frac{68 - 76.6}{\sigma} = -1.645 \Rightarrow \sigma = \frac{8.6}{1.645} \approx \boxed{5.23 \text{ kg}}$$

$$(c) (i) 76.6 \pm 1.5(5.228): \quad 76.6 - 7.84 = 68.76, \quad 76.6 + 7.84 = 84.44$$

$$\boxed{68.8 < W < 84.4}$$

$$(ii) P(68.76 < W < 84.44) = \text{normalcdf}(68.76, 84.44, 76.6, 5.228) \approx \boxed{0.866}$$

(the same as  $P(-1.5 < Z < 1.5)$ )

$$(d) P(\text{woman} \cap T) = 0.25 \times 0.7 = 0.175$$

$$P(\text{woman} | T) = \frac{P(\text{woman} \cap T)}{P(T)} = \frac{0.175}{0.8664} \approx \boxed{0.202}$$

*End of worksheet*

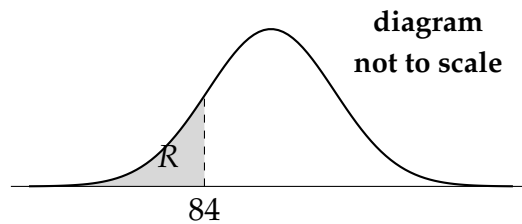
**Question 20**

Non-calculator

Easy

[Maximum mark: 5]

The random variable  $X$  is normally distributed with a mean of 90. The following diagram shows the normal curve for  $X$ .



Let  $R$  be the shaded region under the curve, to the left of 84. The area of  $R$  is 0.22.

- (a) Write down  $P(X < 84)$ . [1]
- (b) Find  $P(84 < X < 90)$ . [2]
- (c) Find  $P(84 < X < 96)$ . [2]

Working:

**Question 21**

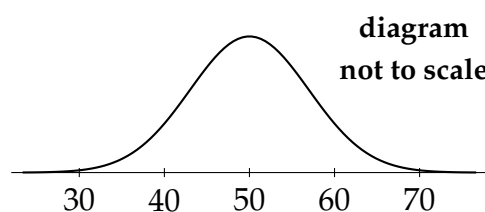
Calculator

Easy

[Maximum mark: 6]

A random variable  $X$  is normally distributed with a mean of 50 and a standard deviation of 10.

- (a) On the following diagram, shade in the region representing  $P(X \geq 45)$ . [2]



- (b) Find  $P(X \geq 45)$ . [2]
- (c) Given that  $P(X \leq k) = 0.8$ , find the value of  $k$ . [2]

Working:

## Answers – Questions 20 & 21

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### Question 20

- (a)  $P(X < 84) = \boxed{0.22}$  (the area of  $R$ )
- (b) By symmetry  $P(X < 90) = 0.5$ , so  $P(84 < X < 90) = 0.5 - 0.22 = \boxed{0.28}$
- (c) 84 and 96 are both 6 from the mean, so  $P(90 < X < 96) = 0.28$  as well.  
 $P(84 < X < 96) = 2 \times 0.28 = \boxed{0.56}$

### Question 21

$X \sim N(50, 10^2)$

- (a) Shade the area under the curve to the **right** of  $x = 45$  (just left of the mean, up to the end of the curve).
- (b)  $P(X \geq 45) = \text{normalcdf}(45, 10^{99}, 50, 10) \approx \boxed{0.691}$
- (c)  $k = \text{invNorm}(0.8, 50, 10) \approx \boxed{58.4}$

**Question 22**

Calculator

Medium

[Maximum mark: 5]

The heights of giraffes in a savanna can be modelled by a normal distribution with mean of 5.2 metres and standard deviation of  $\sqrt{0.1}$  metres.

- (a) Find the probability that a giraffe selected at random is at least 5 metres tall. [2]

Twelve giraffes are independently selected at random and their heights recorded.

- (b) Find the probability that more than nine of these giraffes are at least 5 metres tall. [3]

Working:

**Question 23**

Calculator

Medium

[Maximum mark: 6]

A random variable  $X$  is normally distributed with  $\mu = 120$  and  $\sigma = 15$ .

Find the interquartile range of  $X$ . [6]

Working:

## Answers – Questions 22 & 23

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### Question 22

Let  $H$  be the height of a giraffe.  $H \sim N(5.2, 0.1)$  (variance 0.1, so  $\sigma = \sqrt{0.1} \approx 0.316$ ).

(a)  $P(H \geq 5) = \text{normalcdf}(5, 10^{99}, 5.2, \sqrt{0.1}) \approx \boxed{0.736}$

(b) Let  $Y$  be the number of the 12 giraffes that are at least 5 m tall.  $Y \sim B(12, 0.736455)$ .

$$P(Y > 9) = P(Y \geq 10) = \text{binomCdf}(12, 0.736455, 10, 12) \approx \boxed{0.350}$$

### Question 23

$X \sim N(120, 15^2)$

- $Q_3 = \text{invNorm}(0.75, 120, 15) \approx 130.117$
- $Q_1 = \text{invNorm}(0.25, 120, 15) \approx 109.883$
- $\text{IQR} = Q_3 - Q_1 \approx \boxed{20.2}$

**Question 24**

Calculator

Medium

[Maximum mark: 7]

The time taken for a student to complete an exam is normally distributed with a mean of 40 minutes and a standard deviation of 5.5 minutes.

- (a) A student is selected at random. Find the probability that the student completes the task in less than 48 minutes. [3]
- (b) The probability a student takes between  $k$  and 48 minutes is 0.4. Find the value of  $k$ . [4]

Working:

**Question 25**

Calculator

Hard

[Maximum mark: 6]

The weights  $W$ , in grams, of tea bags are normally distributed with a mean of 3.5 grams and a standard deviation of 0.53 grams. A tea bag is considered small if it weighs less than  $w$  grams.

- (a) Given that 5.2% of tea bags are small, find  $w$ . [3]
- (b) A selected tea bag is small. Find the probability that it weighs at least 2.25 grams. [3]

Working:

## Answers – Questions 24 & 25

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### Question 24

Let  $T$  be the time taken.  $T \sim N(40, 5.5^2)$ .

$$(a) P(T < 48) = \text{normalcdf}(-10^9, 48, 40, 5.5) \approx \boxed{0.927}$$

$$(b) P(k < T < 48) = 0.4 \Rightarrow P(T < k) = 0.927102 - 0.4 = 0.527102$$

$$k = \text{invNorm}(0.527102, 40, 5.5) \approx \boxed{40.4} \text{ minutes}$$

### Question 25

$W \sim N(3.5, 0.53^2)$

$$(a) P(W < w) = 0.052 \Rightarrow w = \text{invNorm}(0.052, 3.5, 0.53) \approx \boxed{2.64} \text{ grams}$$

$$(b) P(W \geq 2.25 \mid W < w) = \frac{P(2.25 \leq W < w)}{P(W < w)} = \frac{0.052 - P(W < 2.25)}{0.052}$$

$$P(W < 2.25) \approx 0.009175, \text{ so } \frac{0.052 - 0.009175}{0.052} \approx \boxed{0.824}$$

**Question 26**

Calculator

Hard

[Maximum mark: 15]

A garden bed has a large number of flowers. The heights of the flowers are normally distributed with a mean of 43 cm and a standard deviation of 6 cm. A flower is classified as tall if it is taller than 48 cm.

- (a) A flower is selected at random.
  - (i) Find the probability that this flower is tall.
  - (ii) Given that the flower is tall, find the probability it is taller than 55 cm. [6]
- (b) Two flowers are selected at random. Find the probability they are both tall. [3]
- (c) Five hundred flowers are selected at random.
  - (i) Find the probability that at least 100 of these 500 flowers are tall.
  - (ii) Find the expected number of tall flowers among these 500 flowers. [6]

Working:

## Answers – Question 26

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### Question 26

Let  $H$  be the height of a flower.  $H \sim N(43, 6^2)$ .

(a) (i)  $P(H > 48) = \text{normalcdf}(48, 10^{99}, 43, 6) \approx \boxed{0.202}$

(ii)  $P(H > 55 \mid H > 48) = \frac{P(H > 55)}{P(H > 48)} = \frac{0.022750}{0.202328} \approx \boxed{0.112}$

(b)  $P(\text{both tall}) = 0.202328^2 \approx \boxed{0.0409}$

(c) Let  $X$  be the number of tall flowers.  $X \sim B(500, 0.202328)$ .

(i)  $P(X \geq 100) = \text{binomCdf}(500, 0.202328, 100, 500) \approx \boxed{0.569}$

(ii)  $E(X) = np = 500 \times 0.202328 \approx \boxed{101}$  flowers

**Question 27**

Calculator

Hard

[Maximum mark: 12]

The weights  $W$ , in grams, of frogs in an enclosure are normally distributed with a mean of 230 grams and a standard deviation of 15 grams.

A frog that weighs less than 210 grams is considered as a small frog. A frog is selected at random.

- (a) Find the probability that this frog is small. [3]

A frog that weighs more than 250 grams is considered as a big frog. Another frog is selected at random.

- (b) Find the probability that the frog is neither small nor big. [3]

Frogs that aren't small or big are considered as standard frogs. A sample of 450 frogs is selected.

- (c) Find the expected number of standard frogs in this sample. [3]

Another sample of  $N$  frogs is selected.

- (d) Find the minimum value of  $N$  such that the expected number of standard frogs in the sample is at least 100. [3]

Working:

## Answers – Question 27

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### Question 27

$$W \sim N(230, 15^2)$$

$$(a) P(W < 210) = \text{normalcdf}(-10^{99}, 210, 230, 15) \approx \boxed{0.0912}$$

$$(b) P(210 \leq W \leq 250) = \text{normalcdf}(210, 250, 230, 15) \approx \boxed{0.818}$$

$$(c) E = 450 \times 0.817578 \approx \boxed{368} \text{ frogs}$$

$$(d) N \times 0.817578 \geq 100 \Rightarrow N \geq 122.3, \text{ so } \boxed{N = 123}$$